## 3.7 THREE-PHASE UNCONTROLLED BRIDGE RECTIFIERS

A three-phase bridge rectifier is commonly used in high-power applications and it is shown in Figure 3.11. This is a *full-wave rectifier*. It can operate with or without a transformer and gives six-pulse ripples on the output voltage. The diodes are numbered in order of conduction sequences and each one conducts for 120°. The conduction sequence for diodes is $D_1 - D_2, D_3 - D_2, D_3 - D_4, D_5 - D_4, D_5 - D_6$, and $D_1 - D_6$. The pair of diodes which are connected between that pair of supply lines having the highest amount of instantaneous line-to-line voltage will conduct. The line-to-line voltage is $\sqrt{3}$ times the phase voltage of a three-phase Y-connected source. The waveforms and conduction times of diodes are shown in Figure 3.12 [4].

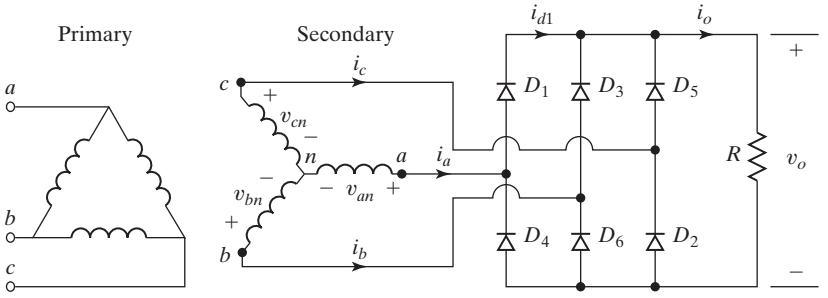



FIGURE 3.11

Three-phase bridge rectifier.

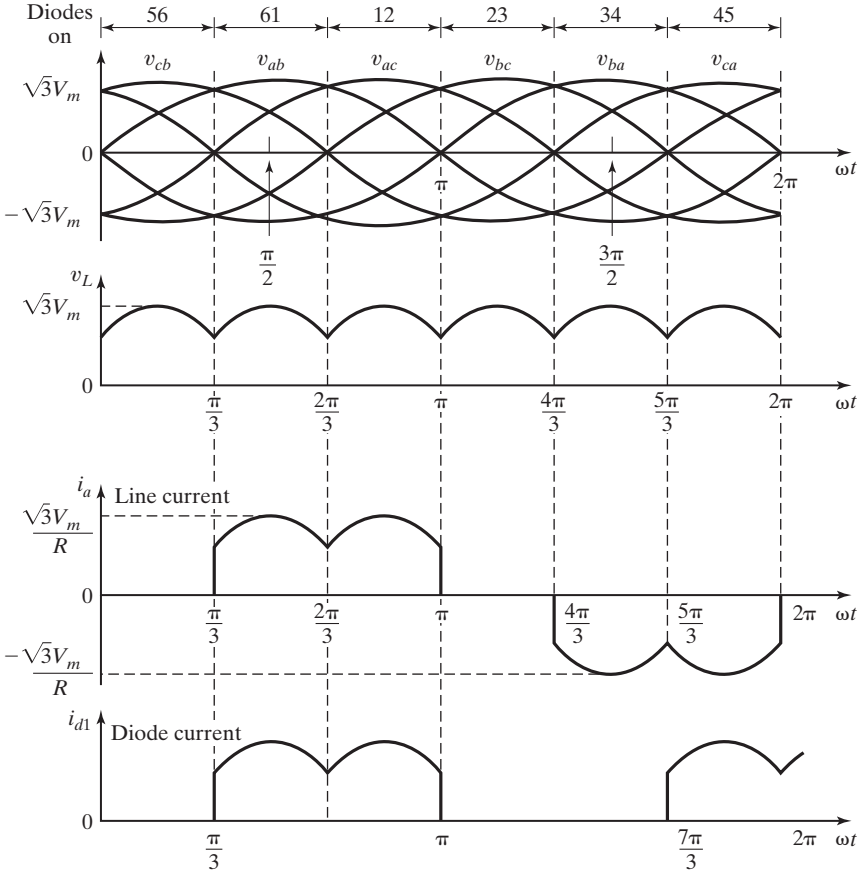



FIGURE 3.12

Waveforms and conduction times of diodes.

If $V_m$ is the peak value of the phase voltage, then the instantaneous phase voltages can be described by

$$v_{an} = V_m \sin(\omega t) \quad v_{bn} = V_{\text{m}} \sin(\omega t - 120°) \quad v_{cn} = V_m \sin(\omega t - 240°)$$

Because the line–line voltage leads the phase voltage by 30°, the instantaneous line–line voltages can be described by

$$v_{ab} = \sqrt{3}\, V_m \sin(\omega t + 30°) \quad v_{bc} = \sqrt{3}\, V_m \sin(\omega t - 90°)$$
$$v_{ca} = \sqrt{3}\, V_m \sin(\omega t - 210°)$$

The average output voltage is found from

$$V_{\text{dc}} = \frac{2}{2\pi/6}\int_0^{\pi/6} \sqrt{3}\, V_m \cos \omega t\, d(\omega t)$$
$$= \frac{3\sqrt{3}}{\pi} V_m = 1.654 V_m \tag{3.33}$$

where $V_m$ is the peak phase voltage. The rms output voltage is

$$V_{\text{rms}} = \left[\frac{2}{2\pi/6}\int_0^{\pi/6} 3V_m^2 \cos^2 \omega t\, d(\omega t)\right]^{1/2}$$
$$= \left(\frac{3}{2} + \frac{9\sqrt{3}}{4\pi}\right)^{1/2} V_m = 1.6554 V_m \tag{3.34}$$

If the load is purely resistive, the peak current through a diode is $I_m = \sqrt{3}\, V_m/R$ and the rms value of the diode current is

$$I_{D(\text{rms})} = \left[\frac{4}{2\pi}\int_0^{\pi/6} I_m^2 \cos^2 \omega t\, d(\omega t)\right]^{1/2}$$
$$= I_m\left[\frac{1}{\pi}\left(\frac{\pi}{6} + \frac{1}{2}\sin\frac{2\pi}{6}\right)\right]^{1/2}$$
$$= 0.5518 I_m \tag{3.35}$$

and the rms value of the transformer secondary current,

$$I_s = \left[\frac{8}{2\pi}\int_0^{\pi/6} I_m^2 \cos^2 \omega t\, d(\omega t)\right]^{1/2}$$
$$= I_m\left[\frac{2}{\pi}\left(\frac{\pi}{6} + \frac{1}{2}\sin\frac{2\pi}{6}\right)\right]^{1/2}$$
$$= 0.7804 I_m \tag{3.36}$$

where $I_m$ is the peak secondary line current.

For a three-phase rectifier $q = 6$, Eq. (3.32) gives the instantaneous output voltage as

$$v_0(t) = 0.9549 V_m\left(1 + \frac{2}{35}\cos(6\omega t) - \frac{2}{143}\cos(12\omega t) + \cdots\right) \tag{3.37}$$

*Note*: In order to increase the number of pulses in the output voltages to 12, two three-phase rectifiers are connected in series. The input to one rectifier is a Y-connected secondary of a transformer and the input to the other rectifier is a delta-connected secondary of a transformer.

## Example 3.7 Finding the Performance Parameters of a Three-Phase Bridge Rectifier

A three-phase bridge rectifier has a purely resistive load of $R$. Determine (a) the efficiency, (b) the FF, (c) the RF, (d) the TUF, (e) the peak inverse (or reverse) voltage (PIV) of each diode, and (f) the peak current through a diode. The rectifier delivers $I_{dc} = 60$ A at an output voltage of $V_{dc} = 280.7$ V and the source frequency is 60 Hz.

***Solution***

**a.** From Eq. (3.33), $V_{dc} = 1.654V_m$ and $I_{dc} = 1.654V_m/R$. From Eq. (3.34), $V_{rms} = 1.6554V_m$ and $I_{o(rms)} = 1.6554V_m/R$. From Eq. (3.1), $P_{dc} = (1.654V_m)^2/R$, from Eq. (3.2), $P_{ac} = (1.654V_m)^2/R$, and from Eq. (3.3) the efficiency

$$\eta = \frac{(1.654V_m)^2}{(1.6554V_m)^2} = 99.83\%$$

**b.** From Eq. (3.5), the FF $= 1.6554/1.654 = 1.0008 = 100.08\%$.

**c.** From Eq. (3.6), the RF $= \sqrt{1.0008^2 - 1} = 0.04 = 4\%$.

**d.** From Eq. (3.15), the rms voltage of the transformer secondary, $V_s = 0.707V_m$. From Eq. (3.36), the rms current of the transformer secondary,

$$I_s = 0.7804I_m = 0.7804 \times \sqrt{3}\,\frac{V_m}{R}$$

The VA rating of the transformer,

$$\text{VA} = 3V_sI_s = 3 \times 0.707V_m \times 0.7804 \times \sqrt{3}\,\frac{V_m}{R}$$

From Eq. (3.8),

$$\text{TUF} = \frac{1.654^2}{3 \times \sqrt{3} \times 0.707 \times 0.7804} = 0.9542$$

The input power factor is

$$\text{PF} = \frac{P_{ac}}{\text{VA}} = \frac{1.6554^2}{3 \times \sqrt{3} \times 0.707 \times 0.7804} = 0.956 \text{ (lagging)}$$

**e.** From Eq. (3.33), the peak line-to-neutral voltage is $V_m = 280.7/1.654 = 169.7$ V. The peak inverse voltage of each diode is equal to the peak value of the secondary line-to-line voltage, PIV $= \sqrt{3}\,V_m = \sqrt{3} \times 169.7 = 293.9$ V.

**f.** The average current through each diode is

$$I_{D(av)} = \ = \frac{4}{2\pi}\int_0^{\pi/6} I_m \cos \omega t\, d(\omega t) = I_m \frac{2}{\pi} \sin \frac{\pi}{6} = 0.3183I_m$$

The average current through each diode is $I_{D(av)} = 60/3 = 20$ A; therefore, the peak current is $I_m = 20/0.3183 = 62.83$ A.

## 10.4 THREE-PHASE CONTROLLED FULL CONVERTERS

Three-phase converters [2, 11] are extensively used in industrial applications up to the 120-kW level, where a two-quadrant operation is required. Figure 10.3a shows a full-converter circuit with a highly inductive load. This circuit is known as a three-phase bridge. The thyristors are turned on at an interval of $\pi/3$. The frequency of output ripple voltage is $6f_s$ and the filtering requirement is less than that of half-wave converters. At $\omega t = \pi/6 + \alpha$, thyristor $T_6$ is already conducting and thyristor $T_1$ is turned on. During interval $(\pi/6 + \alpha) \leq \omega t \leq (\pi/2 + \alpha)$, thyristors $T_1$ and $T_6$ conduct and the line-to-line voltage $v_{ab}(= v_{an} - v_{bn})$ appears across the load. At $\omega t = \pi/2 + \alpha$, thyristor $T_2$ is turned on and thyristor $T_6$ is reverse biased immediately. $T_6$ is turned off due to natural commutation. During interval $(\pi/2 + \alpha) \leq \omega t \leq (5\pi/6 + \alpha)$, thyristors $T_1$ and $T_2$ conduct and the line-to-line voltage $v_{ac}$ appears across the load. If the thyristors are numbered, as shown in Figure 10.3a, the firing sequence is 12, 23, 34, 45, 56, and 61. Figures 10.3b–h show the waveforms for input voltage, output voltage, input current, and currents through thyristors.

If the line-to-neutral voltages are defined as

$$v_{an} = V_m \sin \omega t$$
$$v_{bn} = V_m \sin\left(\omega t - \frac{2\pi}{3}\right)$$
$$v_{cn} = V_m \sin\left(\omega t + \frac{2\pi}{3}\right)$$

the corresponding line-to-line voltages are

$$v_{ab} = v_{an} - v_{bn} = \sqrt{3}\, V_m \sin\left(\omega t + \frac{\pi}{6}\right)$$
$$v_{bc} = v_{bn} - v_{cn} = \sqrt{3}\, V_m \sin\left(\omega t - \frac{\pi}{2}\right)$$
$$v_{ca} = v_{cn} - v_{an} = \sqrt{3}\, V_m \sin\left(\omega t + \frac{\pi}{2}\right)$$

The average output voltage is found from

$$V_{dc} = \frac{3}{\pi}\int_{\pi/6+\alpha}^{\pi/2+\alpha} v_{ab}\, d(\omega t) = \frac{3}{\pi}\int_{\pi/6+\alpha}^{\pi/2+\alpha} \sqrt{3}\, V_m \sin\left(\omega t + \frac{\pi}{6}\right) d(\omega t)$$
$$= \frac{3\sqrt{3}\, V_m}{\pi} \cos \alpha \qquad (10.15)$$

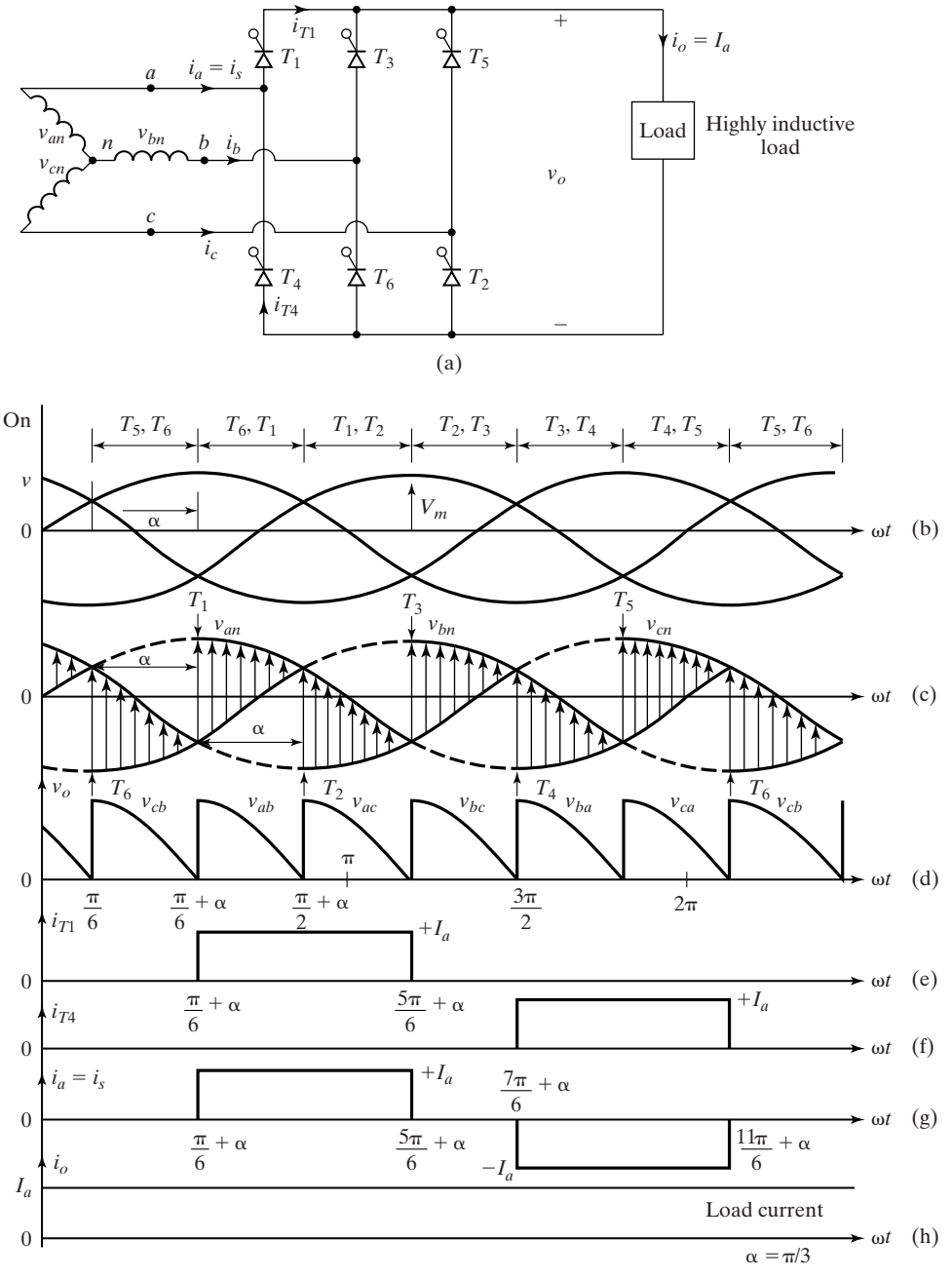



FIGURE 10.3

Three-phase full converter. (a) Circuit, (b) Triggering sequences, (c) Phase voltages, (d) Output voltage (line-to-line voltages), (e) Current through thyristor $T_1$, (f) Current through thyristor $T_2$, (g) Input supply current, and (h) Constant load current.

The maximum average output voltage for delay angle, $\alpha = 0$, is

$$V_{dm} = \frac{3\sqrt{3}\, V_m}{\pi}$$

and the normalized average output voltage is

$$V_n = \frac{V_{\text{dc}}}{V_{dm}} = \cos\alpha \tag{10.16}$$

The rms value of the output voltage is found from

$$V_{\text{rms}} = \left[\frac{3}{\pi}\int_{\pi/6+\alpha}^{\pi/2+\alpha} 3V_m^2 \sin^2\left(\omega t + \frac{\pi}{6}\right) d(\omega t)\right]^{1/2}$$

$$= \sqrt{3}\, V_m\left(\frac{1}{2} + \frac{3\sqrt{3}}{4\pi}\cos 2\alpha\right)^{1/2} \tag{10.17}$$

Figures 10.5b–h show the waveforms for $\alpha = \pi/3$. For $\alpha > \pi/3$, the instantaneous output voltage $v_o$ has a negative part. Because the current through thyristors cannot be negative, the load current is always positive. Thus, with a resistive load, the instantaneous load voltage cannot be negative, and the full converter behaves as a semiconverter.

**Gating sequence.** The gating sequence is as follows:

1. Generate a pulse signal at the positive zero crossing of the phase voltage $v_{an}$. Delay the pulse by the desired angle $\alpha + \pi/6$ and apply it to the gate and cathode terminals of $T_1$ through a gate-isolating circuit.
2. Generate five more pulses each delayed by $\pi/6$ from each other for gating $T_2, T_3, \ldots, T_6$, respectively, through gate-isolating circuits.

## Example 10.4 Finding the Performances of a Three-Phase Full-Wave Converter

A three-phase full-wave converter in Figure 10.3a is operated from a three-phase Y-connected 208-V, 60-Hz supply and the load resistance is $R = 10\,\Omega$. If it is required to obtain an average output voltage of 50% of the maximum possible output voltage, calculate (a) the delay angle $\alpha$, (b) the rms and average output currents, (c) the average and rms thyristor currents, (d) the rectification efficiency, (e) the TUF, and (f) the input PF.

***Solution***

The phase voltage $V_s = 208/\sqrt{3} = 120.1$ V, $V_m = \sqrt{2}\, V_s = 169.83$, $V_n = 0.5$, and $R = 10\,\Omega$. The maximum output voltage $V_{dm} = 3\sqrt{3}V_m/\pi = 3\sqrt{3} \times 169.83/\pi = 280.9$ V. The average output voltage $V_{\text{dc}} = 0.5 \times 280.9 = 140.45$ V.

**a.** From Eq. (10.16), $0.5 = \cos\alpha$, and the delay angle $\alpha = 60°$.

**b.** The average output current $I_{\text{dc}} = V_{\text{dc}}/R = 140.45/10 = 14.05$ A. From Eq. (10.17),

$$V_{\text{rms}} = \sqrt{3} \times 169.83\left[\frac{1}{2} + \frac{3\sqrt{3}}{4\pi}\cos(2 \times 60°)\right]^{1/2} = 159.29\text{ V}$$

and the rms current $I_{\text{rms}} = 159.29/10 = 15.93$ A.

**c.** The average current of a thyristor $I_A = I_{dc}/3 = 14.05/3 = 4.68$ A, and the rms current of a thyristor $I_R = I_{rms}\sqrt{2/6} = 15.93\sqrt{2/6} = 9.2$ A.

**d.** From Eq. (3.3) the rectification efficiency is

$$\eta = \frac{V_{dc}I_{dc}}{V_{rms}I_{rms}} = \frac{140.45 \times 14.05}{159.29 \times 15.93} = 0.778 \quad \text{or} \quad 77.8\%$$

**e.** The rms input line current $I_s = I_{rms}\sqrt{4/6} = 13$ A and the input VAR rating $VI = 3V_sI_s = 3 \times 120.1 \times 13 = 4683.9$ VA. From Eq. (3.8), TUF $= V_{dc}I_{dc}/VI = 140.45 \times 14.05/4683.9 = 0.421$.

**f.** The output power $P_o = I_{rms}^2R = 15.93^2 \times 10 = 2537.6$ W. The PF $= P_o/VI = 2537.6/4683.9 = 0.542$ (lagging).

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*Note*: The PF is less than that of three-phase semiconverters, but higher than that of three-phase half-wave converters.

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## Example 10.5 Finding the Input Power Factor of a Three-Phase Full Converter

The load current of a three-phase full converter in Figure 10.3a is continuous with a negligible ripple content. (a) Express the input current in Fourier series, and determine the HF of input current, the DF, and the input PF. (b) If the delay angle $\alpha = \pi/3$, calculate $V_n$, HF, DF, and PF.

### *Solution*

**a.** The waveform for input current is shown in Figure 10.3g and the instantaneous input current of a phase can be expressed in a Fourier series as

$$i_s(t) = a_0 + \sum_{n=1,2,\ldots}^{\infty} (a_n \cos n\omega t + b_n \sin n\omega t)$$

where

$$a_o = \frac{1}{2\pi}\int_0^{2\pi} i_s(t)\, d(\omega t) = 0$$

$$a_n = \frac{1}{\pi}\int_0^{2\pi} i_s(t) \cos n\omega t\, d(\omega t)$$

$$= \frac{1}{\pi}\left[\int_{\pi/6+\alpha}^{5\pi/6+\alpha} I_a \cos n\omega t\, d(\omega t) - \int_{7\pi/6+\alpha}^{11\pi/6+\alpha} I_\alpha \cos n\omega t\, d(\omega t)\right]$$

$$= -\frac{4I_a}{n\pi}\sin\frac{n\pi}{3}\sin n\alpha \quad \text{for } n = 1, 3, 5, \ldots$$

$$= 0 \quad \text{for } n = 2, 4, 6, \ldots$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} i_s(t) \sin n\omega t\, d(\omega t)$$

$$= \frac{1}{\pi}\left[\int_{\pi/6+\alpha}^{5\pi/6+\alpha} I_a \sin n\omega t\, d(\omega t) - \int_{7\pi/6+\alpha}^{11\pi/6+\alpha} I_a \sin n\omega t\, d(\omega t)\right]$$

$$= \frac{4I_a}{n\pi}\cos\frac{n\pi}{6}\cos n\alpha \quad \text{for } n = 1, 3, 5, \ldots$$
$$= 0 \quad \text{for } n = 2, 4, 6, \ldots$$

Because $a_0 = 0$ and the triplen harmonic currents (for $n$ = multiple of 3) will be absent in a balanced three-phase supply, the input current can be written as

$$i_s(t) = \sum_{n=1,3,5,\ldots}^{\infty} \sqrt{2}\, I_{sn} \sin(n\omega t + \phi_n) \quad \text{for } n = 1, 5, 7, 11, 13, \ldots$$

where

$$\phi_n = \tan^{-1}\frac{a_n}{b_n} = -n\alpha \tag{10.18}$$

The rms value of the $n$th harmonic input current is given by

$$I_{sn} = \frac{1}{\sqrt{2}}(a_n^2 + b_n^2)^{1/2} = \frac{2\sqrt{2}\, I_a}{n\pi}\sin\frac{n\pi}{3} \tag{10.19}$$

The rms value of the fundamental current is

$$I_{s1} = \frac{\sqrt{6}}{\pi} I_a = 0.7797 I_a$$

The rms input current

$$I_s = \left[\frac{2}{2\pi}\int_{\pi/6+\alpha}^{5\pi/6+\alpha} I_a^2\, d(\omega t)\right]^{1/2} = I_a\sqrt{\frac{2}{3}} = 0.8165 I_a$$
$$\text{HF} = \left[\left(\frac{I_s}{I_{s1}}\right)^2 - 1\right]^{1/2} = \left[\left(\frac{\pi}{3}\right)^2 - 1\right]^{1/2} = 0.3108 \quad \text{or} \quad 31.08\%$$
$$\text{DF} = \cos\phi_1 = \cos(-\alpha)$$
$$\text{PF} = \frac{I_{s1}}{I_s}\cos(-\alpha) = \frac{3}{\pi}\cos\alpha = 0.9549\ \text{DF}$$

**b.** For $\alpha = \pi/3, V_n = \cos(\pi/3) = 0.5$ pu, HF = 31.08%, DF = $\cos 60° = 0.5$, and PF = 0.478 (lagging).

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*Note*: We can notice that the input PF depends on the delay angle $\alpha$.